

Orjun Kumar Paul

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PROFESSIONAL SUMMARY:

Electrical and Electronic Engineering graduate from Bangladesh University of Engineering and Technology (major in Communication and Signal Processing) with practical exposure to telecommunication systems, network operations, embedded systems, IoT, AI/ML, computer vision and signal processing. Authored a novel frequency-domain deep learning architecture for the undergraduate thesis, and delivered 12 hands-on projects spanning edge AI, robotics, IoT, communication circuits and power systems. Strong foundation in communication systems, programming, technical documentation and problem-solving, with a keen interest in ICT, telecom and intelligent digital engineering solutions.

EDUCATION:

- Bangladesh University of Engineering and Technology (BUET)** 2026
Bachelor of Science in Electrical and Electronic Engineering (Major: Communication and Signal Processing)
CGPA: 3.57/4.00
- Govt. M.M. College, Jashore** 2020
Higher Secondary Certificate (HSC), Science
GPA: 5.00/5.00
- Shaheed Smarani Jhanpa M.L. High School** 2018
Secondary School Certificate (SSC), Science
GPA: 5.00/5.00

WORK EXPERIENCE:

- Bangladesh Telecommunications Company Limited (BTCL) (Industrial Attachment)** November 2025
 - Gained hands-on exposure to telecommunication systems, network operations and industry-level engineering practices.
- IEEE Robotics and Automation Society (RAS), BUET Student Branch Chapter** 2024-Present
 - Served as General Secretary, coordinating chapter activities, events and member engagement.
- BUET Rover Scout Group**
 - Social Volunteer.

TECHNICAL SKILLS:

- Programming:** Python, C++, MATLAB, SystemVerilog
- AI/ML and Computer Vision:** TensorFlow, CNNs, attention mechanisms, LSTM/GRU, OpenCV, YOLO.
- Embedded Systems and IoT:** STM32, ESP32, ESP8266, Raspberry Pi, Arduino, Blynk, sensor interfacing, cloud-based monitoring
- Communication and Signal Processing:** AM/DSB modulation and demodulation, filter design, FFT/PSD spectral analysis, random signal processing
- Engineering Tools:** MATLAB/Simulink, PSpice, Proteus, AutoCAD, PCB design
- Core Areas:** Communication systems, wireless and optical communication, satellite and radar communication, control systems, power electronics, robotics and automation
- Professional Skills:** Technical documentation, teamwork, leadership, presentation, problem-solving, Microsoft Office

RESEARCH EXPERIENCE AND PUBLICATIONS:

- Frequency Channel Adaptive Network (FCAN) for Silicon Wafer Defect Detection and Classification** 2025-2026
 - Undergraduate thesis supervised by Dr. Mohammed Imamul Hassan Bhuiyan, Professor, EEE, BUET.
 - Designed FCAN, a novel deep learning architecture on a ResNet50V2 backbone with three original modules: Spectral Feature Learning (2D FFT and radial Power Spectral Density), Frequency-Guided Cross-Attention (multi-head) and Multi-Scale Feature Fusion (dilated convolutions).
 - Proposed a custom Adaptive Multi-Label Focal Loss (AMFL) for severe class imbalance, and engineered a mixed-defect synthesis paradigm that expanded training data and improved generalization.
 - Achieved AUC-ROC of 0.9991 and Macro F1 of 0.9916 on WM-811K and MixedWM38, outperforming 14 published baseline methods.

LANGUAGES:

- Bengali:** Native
- English:** Professional working proficiency (Academic)

AREAS OF INTEREST:

- Telecommunications and ICT Systems | Deep Learning and Computer Vision | Digital Signal Processing and Communication Systems | Embedded Systems, IoT and Intelligent Assistive Systems | Industrial Automation and Smart Manufacturing | Semiconductor Defect Classification and Pattern Recognition

PROJECTS:

- Intelligent Personal Assistive Robot for Emergency Awareness and Communication Support** 2026
 - Built a Bangla sign-language-to-voice and voice-to-text robot for hearing and speech impaired users, using an INT8-quantized MobileNetV2 (~610 KB) trained on 1,200 custom images; 100% test accuracy, under 2 s gesture-to-audio latency, fire-gesture emergency alert, total cost 18,000 BDT.
 - My contribution:** Core member of a 5-person team; worked on system design, implementation and testing of the integrated hardware and AI pipeline.
 - Tools: Raspberry Pi Zero 2W, ESP32-S3, TensorFlow Lite, MobileNetV2, NLP, Mecanum wheels

- **Electrical Services Design of an 11-Storied High-Rise Residential Building** 2026
 - Complete design for a basement, ground and 9 typical floors: conduit layouts, switchboard diagrams, load schedules, generator/transformer ratings and lightning protection.
 - **My contribution:** Member of a 7-person team contributing to layouts, load calculations and drawings.
 - Tools: AutoCAD, load calculation, protection design
- **Stock Market Prediction with Sentiment Analysis** 2025
 - Predicted Google's closing price by fusing daily news sentiment with technical indicators (MA, MACD, RSI, Bollinger Bands); benchmarked LSTM, GRU, ANN and regressors, with the best GRU model reaching R^2 of 0.9515.
 - **My contribution:** Collected and aligned the datasets and built the sentiment pipeline (text cleaning, polarity scoring, daily aggregation).
 - Tools: Python, LSTM, GRU, XGBoost, Random Forest, TextBlob, VADER
- **IoT-Based Smart Poultry Farm Automation System** 2025
 - Monitored temperature, humidity, gas, light, water and feed levels with automatic control of heater, fan, humidifier, pump and feeder, plus remote monitoring and manual override through a mobile app.
 - **My contribution:** Hardware design, circuit implementation and debugging.
 - Tools: ESP32, Blynk IoT, DHT22, MQ135, ultrasonic and load-cell sensors, relays
- **Fire Fighter Robot Based on Image Processing** 2024
 - Autonomous robot that detects flames with a YOLO model, steers toward the fire using PID control and extinguishes it with a servo-aimed water pump.
 - **My contribution:** Prepared the chassis, assembled and wired the robot, built the water-pump section, debugged code and demonstrated the system.
 - Tools: YOLO, OpenCV, Python, Arduino, PID control
- **BLDC Motor Driver (DC to 3-Phase AC Conversion)** 2024
 - Six-step commutation driver with a three-phase MOSFET inverter, opto-isolated gate drivers, a discrete 5 V regulator and potentiometer speed control.
 - **My contribution:** Proteus schematic design, motor-driver circuit implementation and regulator debugging.
 - Tools: Arduino, IRF3205 MOSFETs, TLP250 optocouplers, Proteus
- **Simulink-Based Program for Power System Dynamic Analysis** 2024
 - Load-flow and overcurrent-relay study of the IEEE 9-bus system under LG, LL, LLG and LLLG faults, plus MATLAB models of generator, excitation and turbine dynamics.
 - **My contribution:** Prepared the MATLAB code for the dynamic models and transient response analysis.
 - Tools: MATLAB, Simulink, IEEE 9-bus system
- **Digital Attendance System by Face Detection** 2024
 - Real-time webcam face recognition that logs name, date and time to Excel with optional email notification.
 - **My contribution:** Coding, face-recognition module and debugging.
 - Tools: MATLAB, HOG features, SVM
- **IoT-Based Smart Door Lock System** 2024
 - Developed a Wi-Fi door lock with remote lock/unlock, access logging and real-time notifications via a cloud database.
 - **My contribution:** Designed the PCB and wrote firmware.
 - Tools: NodeMCU ESP8266, cloud database, Proteus
- **8-Player Quiz Buzzer System** 2024
 - Microcontroller-free buzzer using a priority encoder and latches for first-responder lock-out, per-player scoring and a 9-second countdown timer.
 - **My contribution:** Team member (4-person) in circuit design, implementation and testing.
 - Tools: 74-series logic ICs, 555 timer, 7-segment displays
- **Amplitude Modulation and Demodulation System** 2023
 - Built a DSB amplitude modulation/demodulation chain with a quadrature-oscillator carrier, diode-ring four-quadrant multiplier, narrow band-pass filters (100/300/500 Hz) and a Butterworth low-pass filter, verified in time and frequency domain.
 - **My contribution:** Co-designed and simulated all sub-circuits (2-member team) and analyzed the spectrum at each stage.
 - Tools: PSpice, operational amplifiers, analog filter design
- **Heat Conduction Simulation** 2023
 - GUI-based solver for 1D/2D/3D steady-state heat conduction using the finite difference method.
 - **My contribution:** Co-developed the solver and GUI (2-member team).
 - Tools: MATLAB App Designer, Finite Difference Method

AWARDS AND ACHIEVEMENTS:

- BUET Technical Scholarship
- Talent Pool Scholarship (Class VIII)

REFERENCE:

- **Dr. Mohammed Imamul Hassan Bhuiyan | Thesis Supervisor**
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 Bangladesh University of Engineering and Technology
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